

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_1204.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-06 16:45:00

Training Start: 2025-05-06 15-07-10

Training End: 2025-05-06 15-11-43

Training Duration: 4.55 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.4877

MSE: 3.6826

RMSE: 1.9190

R2 Score: 0.7839

Multi-Step Forecast:

Forecast Steps: 20

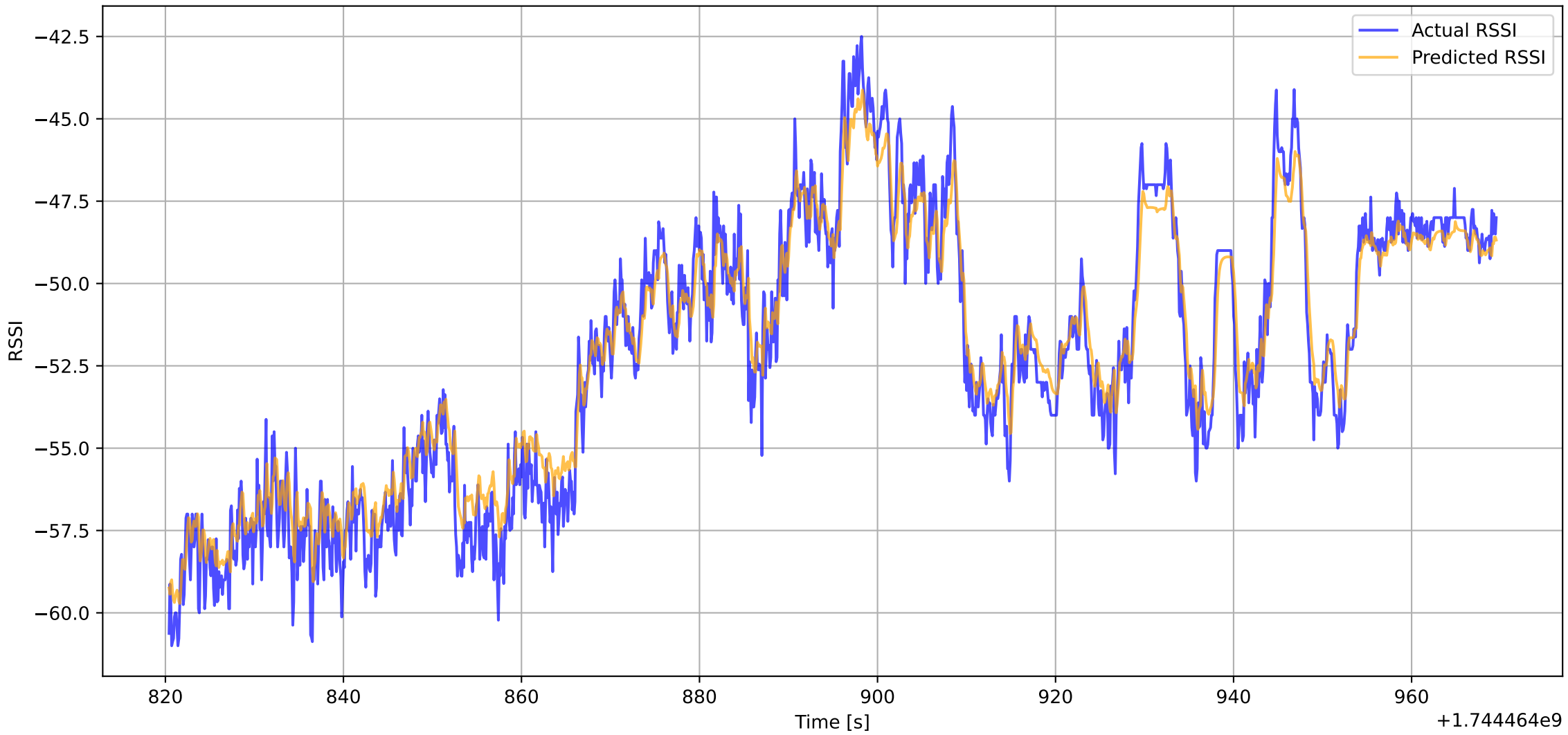
MAE: 0.0303

MSE: 0.0015

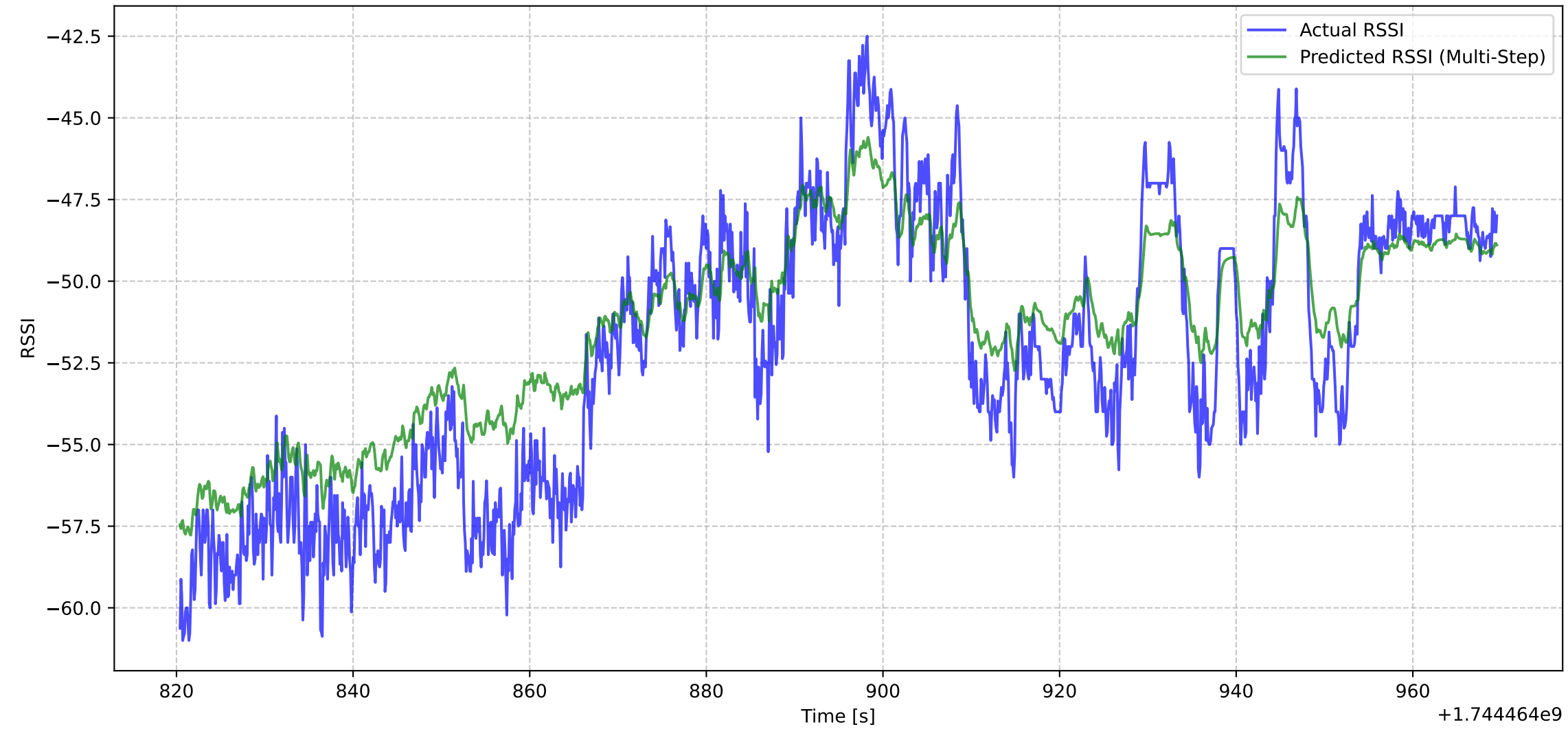
RMSE: 0.0381

R2 Score: 0.9384

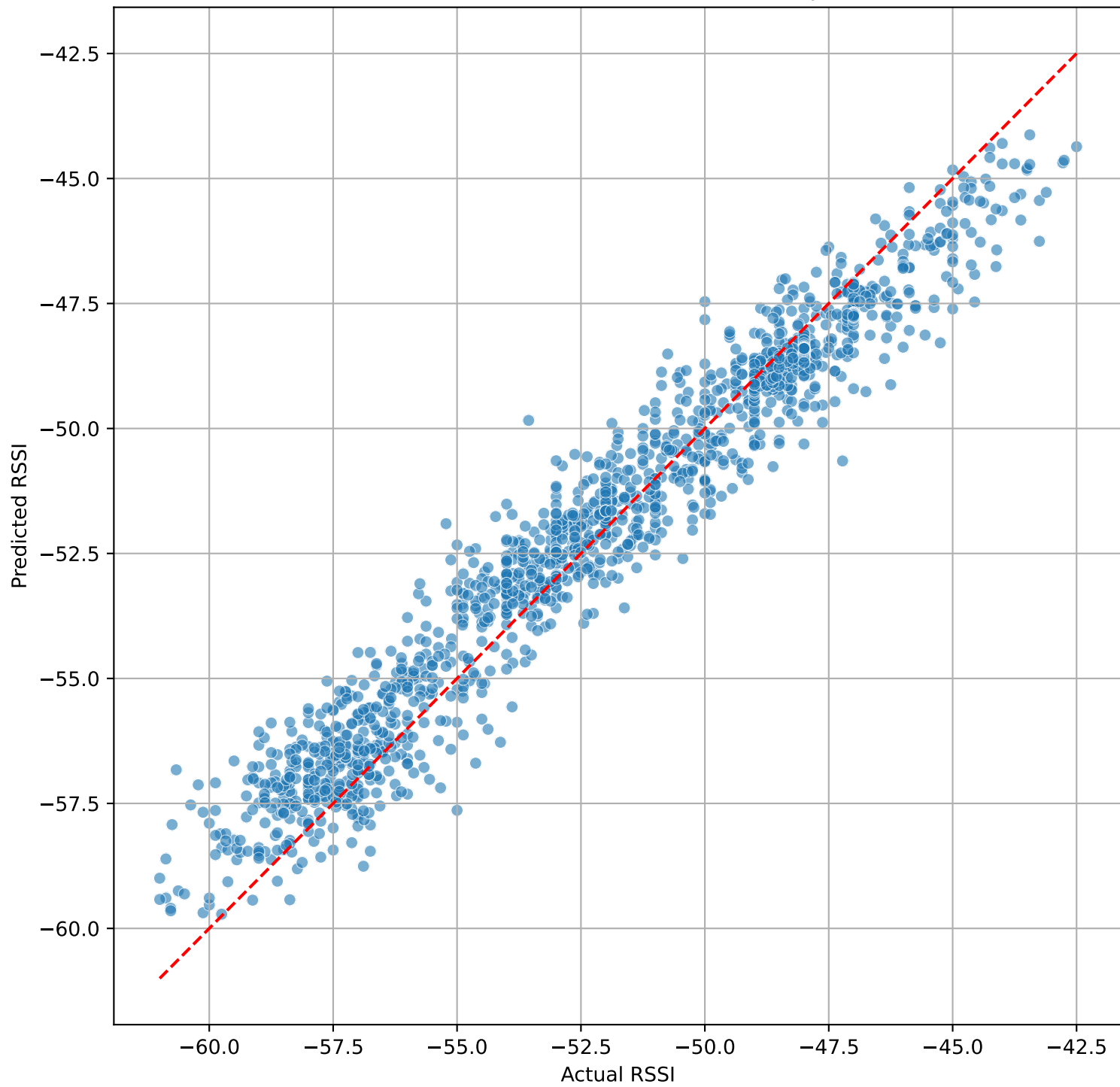
Actual vs Predicted RSSI (First Step)



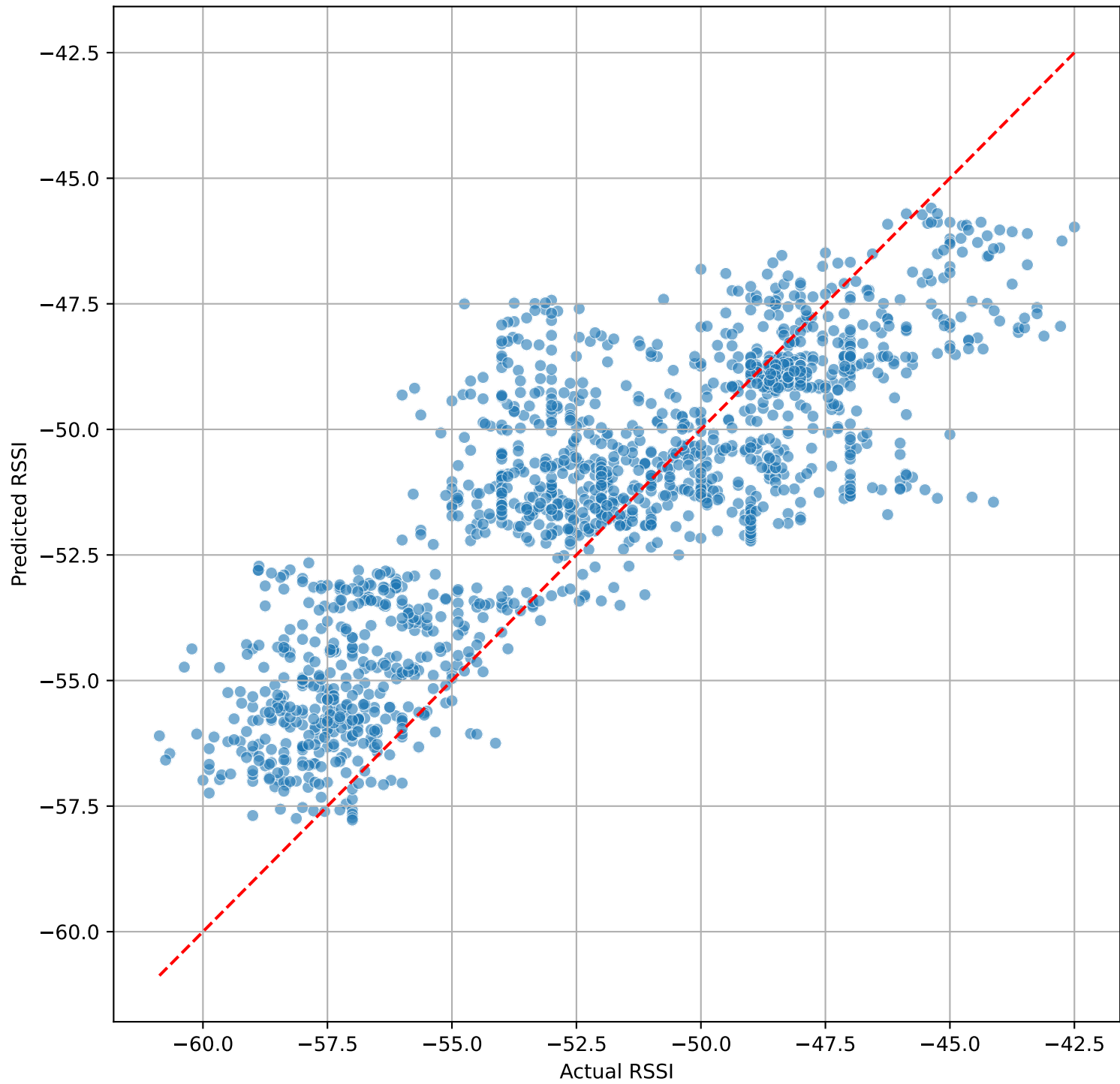
Actual vs Predicted RSSI (Multi-Step (step 20))



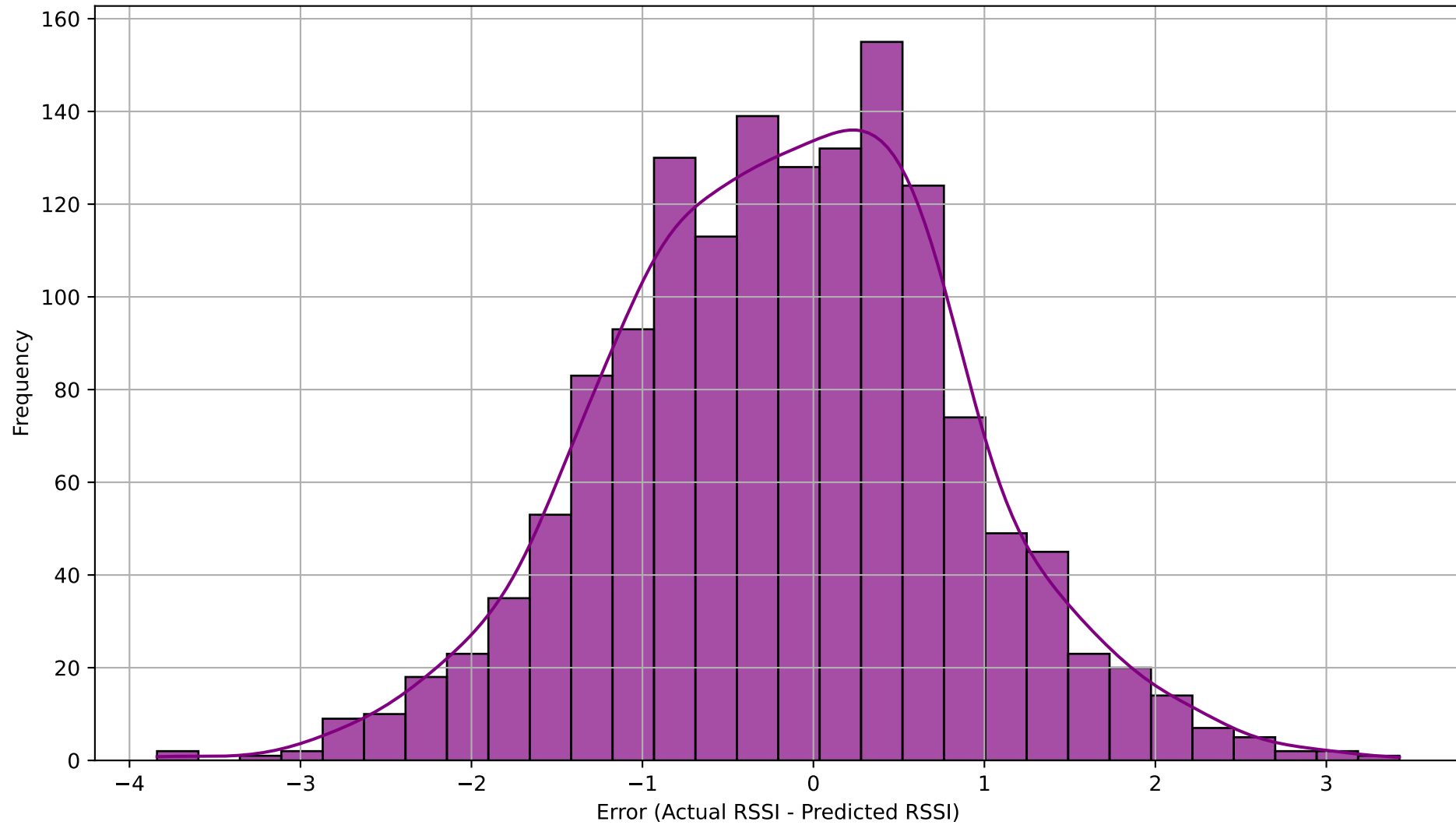
Actual vs Predicted RSSI (First Step)



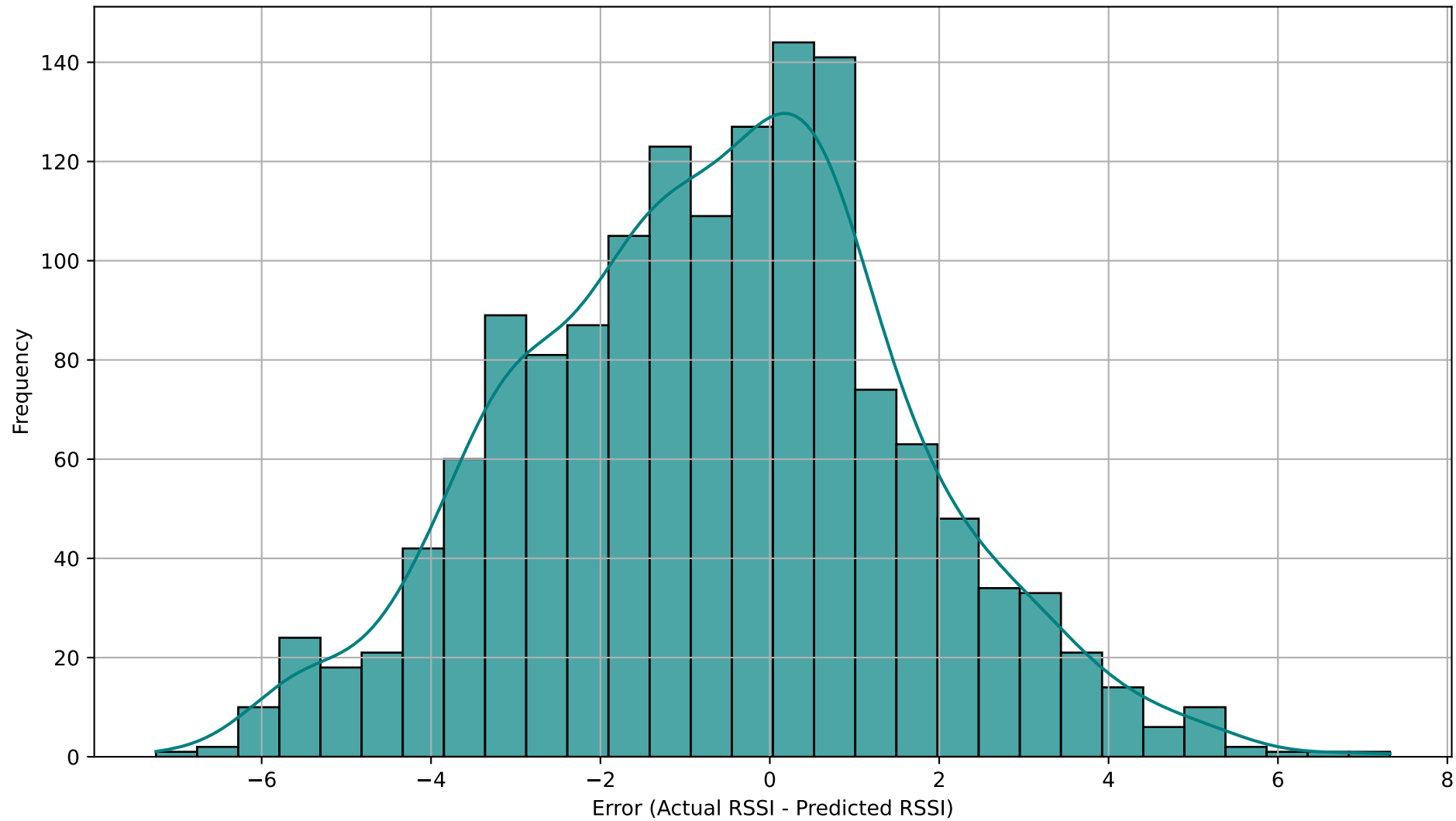
Actual vs Predicted RSSI (Last Step - step 20)



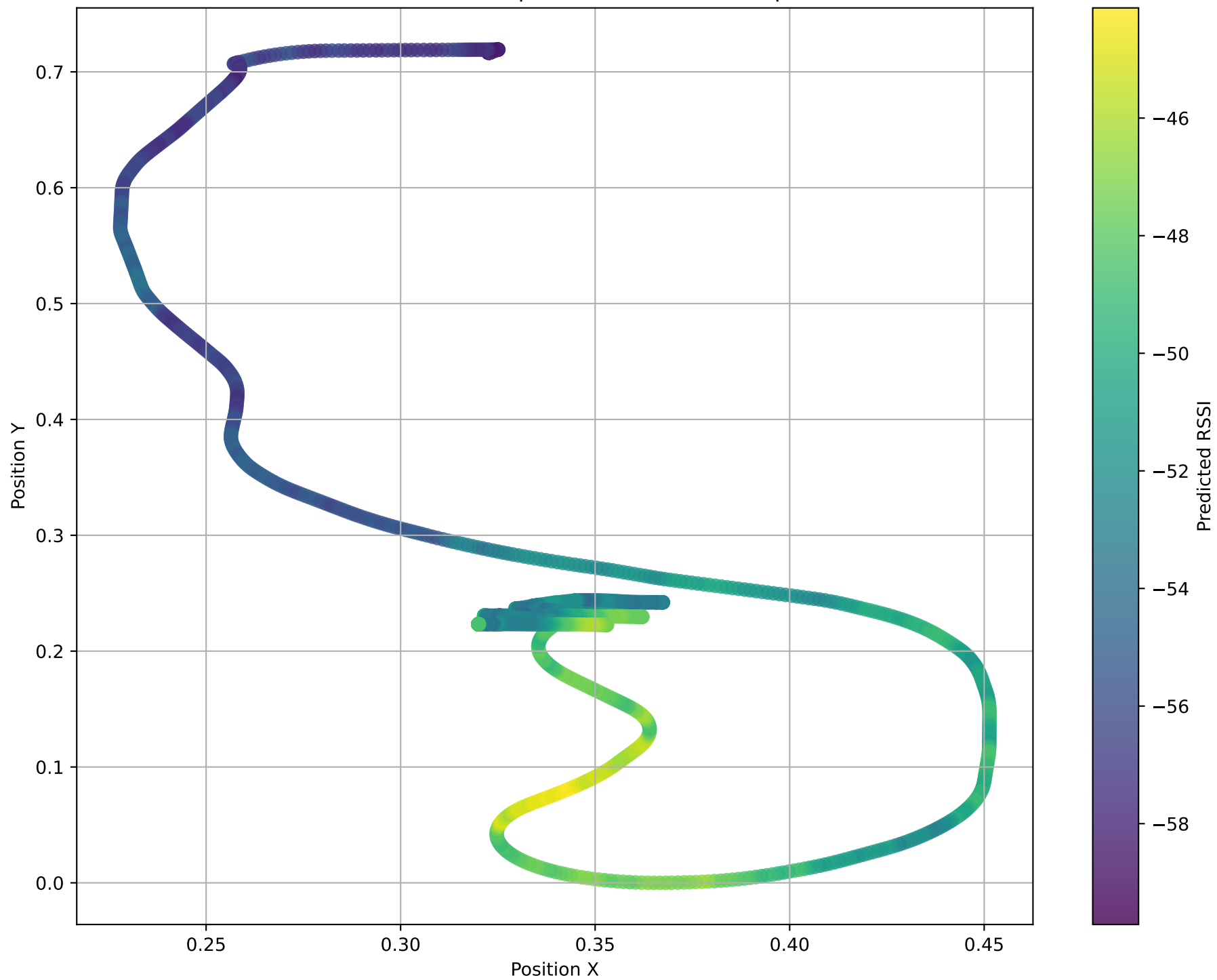
Distribution of Prediction Errors (First Step)



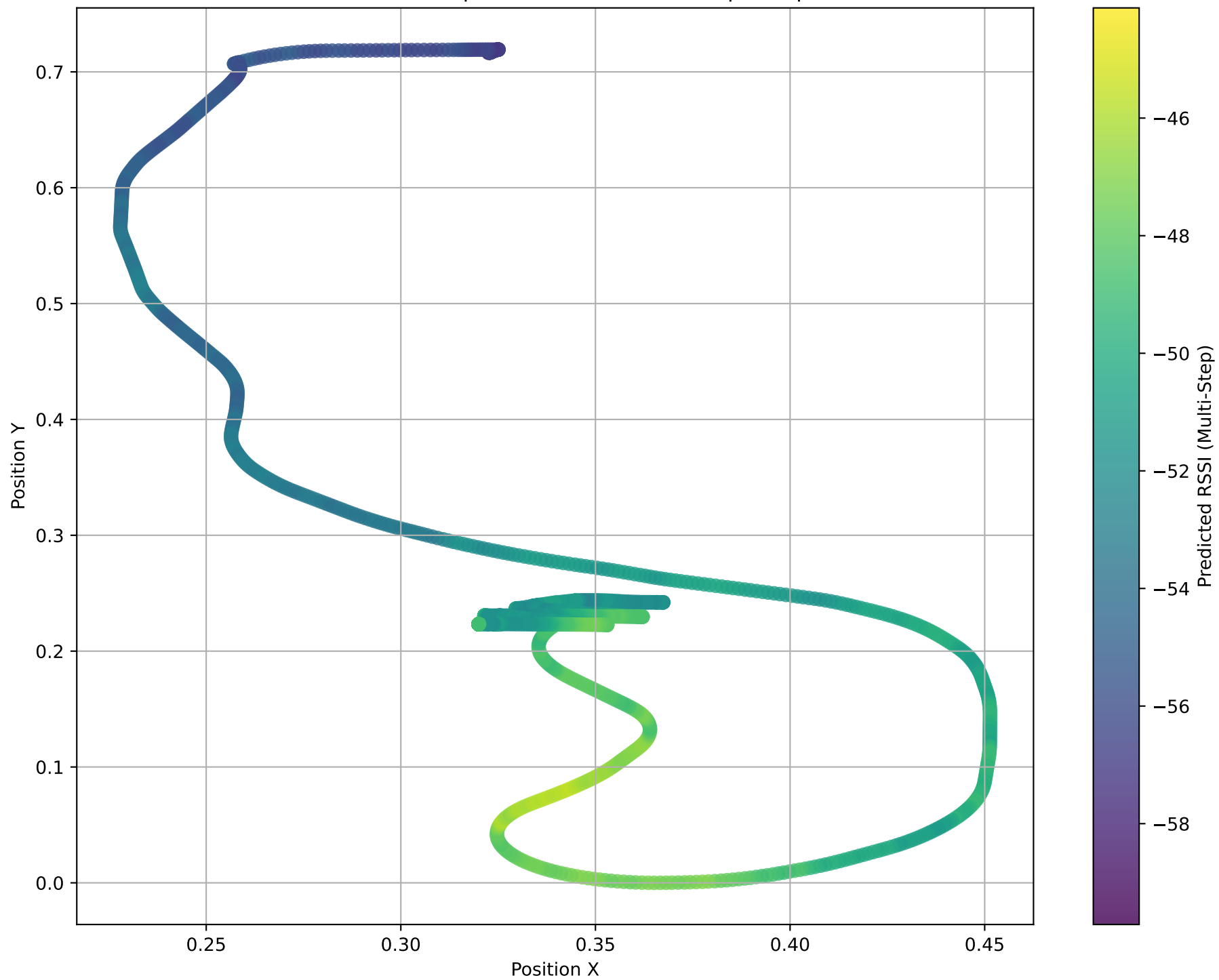
Distribution of Prediction Errors (Last Step - step 20)



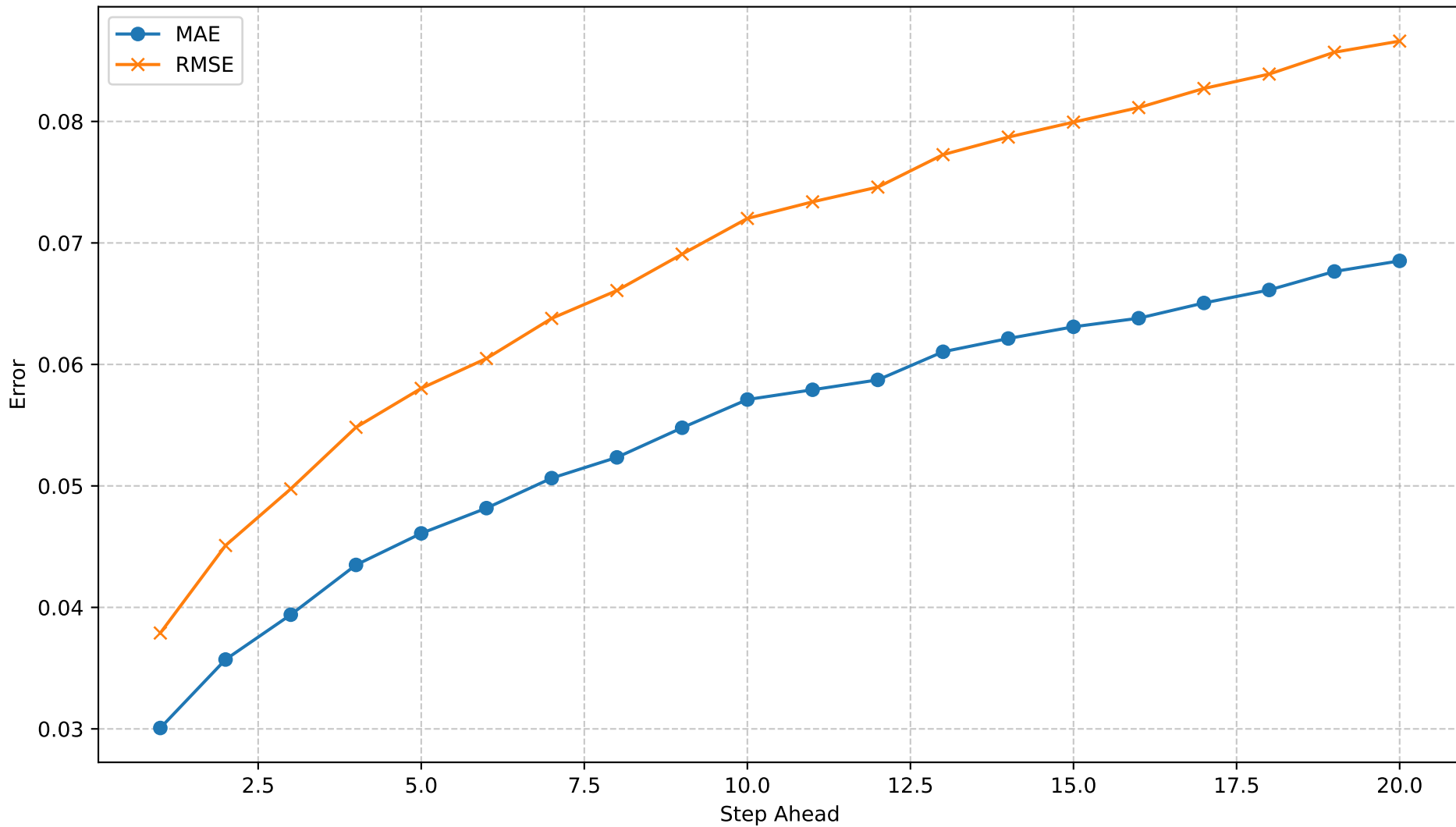
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_1204.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.06 / 0.07
- 20º passo : 0.07 / 0.09

Crescimento do erro: 127.8% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.07), 19 (0.07), 18 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.